DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Introduction to Electrical and Electronics Engineering | Course Code: 25EN1109 | Credits: 4

Exam Session: Mid Semester Exam (CIA-1) March 2026 (Max Marks: 40, Duration: 75m) Max Marks: 80 | Duration: 3 Hours

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INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Calculate the total energy consumed in kWh in a 30-day month for a residence running 2 bulbs of 100 W each and 2 ceiling fans of 60 W each for 10 hours daily. Find the electricity bill if cost is Rs 2.00 per unit. **[6 Marks]**

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Q2. Derive the closed-loop voltage gain expression for an Op-Amp in: (i) Inverting Amplifier configuration, (ii) Non-Inverting Amplifier configuration, and (iii) Summing Amplifier. State 4 characteristics of an ideal Op-Amp. **[8 Marks]**

MODULE 2

Q3. A 440 kVA, 6600V/440V, 50Hz single-phase transformer has 1200 primary turns. Calculate: (i) Number of secondary turns, (ii) Primary full-load current, (iii) Secondary full-load current, and (iv) Maximum magnetic flux in core. **[8 Marks]**

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Q4. Show how NAND and NOR gates serve as Universal Gates by synthesizing AND, OR, NOT, and XOR gates using NAND gates only. **[8 Marks]**